

ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. GATE CE-1 2026

A. Formulae

1.1.1 Intersection of Conics and Common Chord

- Conic in Vector Form: Every second-degree curve $ax^2 + 2bxy + cy^2 + 2dx + 2ey + f = 0$ can be written in matrix-vector form as[cite: 1]:

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0, \quad \mathbf{V} = \begin{pmatrix} a & b \\ b & c \end{pmatrix} = \mathbf{V}^\top, \quad \mathbf{u} = \begin{pmatrix} d \\ e \end{pmatrix} \quad (1.1.1.1)$$

The conic is a parabola if $|\mathbf{V}| = 0$, an ellipse if $|\mathbf{V}| > 0$, and a hyperbola if $|\mathbf{V}| < 0$ [cite: 1]. For a circle with centre $\mathbf{c}$ and radius r , $\|\mathbf{x} - \mathbf{c}\|^2 = r^2$ gives $\mathbf{V} = \mathbf{I}$, $\mathbf{u} = -\mathbf{c}$, and $f = \|\mathbf{c}\|^2 - r^2$ [cite: 1].

- Line-Conic Intersection: The points of intersection of a line $L : \mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ ($\kappa \in \mathbb{R}$) with a conic section $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$ are given by[cite: 1]:

$$\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m} \quad (1.1.1.2)$$

where the parameter values κ_i are[cite: 1]:

$$\kappa_i = \frac{-\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h}) (\mathbf{m}^\top \mathbf{V} \mathbf{m})}}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \quad (1.1.1.3)$$

with $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V} \mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f$ [cite: 1]. The line meets the conic at two, one, or no real points according as the discriminant $\Delta = [\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h}) (\mathbf{m}^\top \mathbf{V} \mathbf{m})$ is positive, zero, or negative[cite: 1].

- Derivation of the Intersection Parameter: Substituting $\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ into the conic equation and using symmetry $\mathbf{V} = \mathbf{V}^\top$ [cite: 1]:

$$(\mathbf{h} + \kappa \mathbf{m})^\top \mathbf{V} (\mathbf{h} + \kappa \mathbf{m}) + 2\mathbf{u}^\top (\mathbf{h} + \kappa \mathbf{m}) + f = 0 \quad (1.1.1.4)$$

$$\kappa^2 (\mathbf{m}^\top \mathbf{V} \mathbf{m}) + 2\kappa \mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) + g(\mathbf{h}) = 0 \quad (1.1.1.5)$$

When $\Delta = 0$, the two roots coincide and the line is tangent to the conic[cite: 1].

- Pencil of Conics: The intersection of two conics with parameters $(\mathbf{V}_1, \mathbf{u}_1, f_1)$ and $(\mathbf{V}_2, \mathbf{u}_2, f_2)$ lies on every member of the pencil[cite: 1]:

$$\mathbf{x}^\top (\mathbf{V}_1 + \mu \mathbf{V}_2) \mathbf{x} + 2(\mathbf{u}_1 + \mu \mathbf{u}_2)^\top \mathbf{x} + (f_1 + \mu f_2) = 0 \quad (1.1.1.6)$$

- Degenerate Conic and Common Chord: The pencil represents a pair of straight lines (a degenerate conic) if and only if[cite: 1]:

$$\begin{vmatrix} \mathbf{V}_1 + \mu \mathbf{V}_2 & \mathbf{u}_1 + \mu \mathbf{u}_2 \\ (\mathbf{u}_1 + \mu \mathbf{u}_2)^\top & f_1 + \mu f_2 \end{vmatrix} = 0 \quad (1.1.1.7)$$

If $\mathbf{V}_1 = \mathbf{V}_2$ (e.g., two circles), choosing $\mu = -1$ removes the second-degree terms, leaving the common chord (radical axis)[cite: 1]:

$$2(\mathbf{u}_1 - \mathbf{u}_2)^\top \mathbf{x} + (f_1 - f_2) = 0 \quad (1.1.1.8)$$

1.1.2 Line Intersection and Angle Between Vectors

- Forms of a Line: A line through point $\mathbf{A}$ with direction vector $\mathbf{m}$ can be written in parametric form or normal form using normal vector $\mathbf{n}$ ($\mathbf{n}^\top \mathbf{m} = 0$)[cite: 1]:

$$\mathbf{x} = \mathbf{A} + \kappa \mathbf{m}, \quad \mathbf{n}^\top (\mathbf{x} - \mathbf{A}) = 0 \quad (1.1.2.1)$$

In 2D, if $\mathbf{m} = \begin{pmatrix} m_1 \\ m_2 \end{pmatrix}$, then $\mathbf{n} = \begin{pmatrix} -m_2 \\ m_1 \end{pmatrix}$ [cite: 1].

- Line Intersection and Solvability: The lines $\mathbf{x} = \mathbf{A} + \kappa_1 \mathbf{m}_1$ and $\mathbf{x} = \mathbf{B} + \kappa_2 \mathbf{m}_2$ meet where[cite: 1]:

$$\begin{pmatrix} \mathbf{m}_1 & -\mathbf{m}_2 \end{pmatrix} \begin{pmatrix} \kappa_1 \\ \kappa_2 \end{pmatrix} = \mathbf{B} - \mathbf{A} \quad (1.1.2.2)$$

This system has a unique solution if and only if $|\mathbf{m}_1 \quad \mathbf{m}_2| \neq 0$ [cite: 1]. If the determinant is zero, the lines are parallel or coincident[cite: 1].

- Angle Between Vectors: The angle θ between two vectors $\mathbf{a}$ and $\mathbf{b}$ is given by[cite: 1]:

$$\cos \theta = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \quad (1.1.2.3)$$

Two vectors are orthogonal if and only if $\mathbf{a}^\top \mathbf{b} = 0$ [cite: 1].

- Circle Representation and Chords: A point on a circle of radius r centered at the origin is $\mathbf{x} = r \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix}$ with $\mathbf{e}_\alpha \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix}$ [cite: 1]. The chord length between $\mathbf{x}_\alpha = r \mathbf{e}_\alpha$ and $\mathbf{x}_\beta = r \mathbf{e}_\beta$ satisfies[cite: 1]:

$$\|\mathbf{x}_\alpha - \mathbf{x}_\beta\|^2 = 2r^2 (1 - \mathbf{e}_\alpha^\top \mathbf{e}_\beta) = 4r^2 \sin^2 \frac{\alpha - \beta}{2} \quad (1.1.2.4)$$

1.1.3 System of Linear Equations and Rouché–Capelli Theorem

- Consistency and Solvability: A system $\mathbf{A}\mathbf{x} = \mathbf{b}$ with $\mathbf{A} \in \mathbb{R}^{m \times n}$ is consistent if and only if[cite: 1]:

$$\text{rank}(\mathbf{A}) = \text{rank}([\mathbf{A} \mid \mathbf{b}]) = r \quad (1.1.3.1)$$

If $r = n$, the solution is unique; if $r < n$, there exist infinitely many solutions with $n - r$ free variables[cite: 1]. If $\mathbf{A} \in \mathbb{R}^{n \times (n-1)}$, the system is consistent only if $|\mathbf{A} \quad \mathbf{b}| = 0$ [cite: 1].

- Homogeneous System and Zero Norm Property: For $\mathbf{A} \in \mathbb{R}^{2 \times 3}$ with rank 2, the solution of $\mathbf{A}\mathbf{x} = \mathbf{0}$ is given by the cross product of its rows[cite: 1]:

$$\mathbf{x} = \kappa (\mathbf{a}_1 \times \mathbf{a}_2) \quad (1.1.3.2)$$

For any vector $\mathbf{v} \in \mathbb{R}^n$, $\|\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{v} = \sum_i v_i^2 = 0 \iff \mathbf{v} = \mathbf{0}$ [cite: 1].

1.1.4 Relation Between Eigenvalues and Trace of a Matrix

- Characteristic Polynomial and Trace Relation: The characteristic polynomial of $\mathbf{A} \in \mathbb{R}^{n \times n}$ is $P(\lambda) = |\lambda \mathbf{I} - \mathbf{A}|$ [cite: 1]. The sum of all eigenvalues equals the

trace of the matrix[cite: 1]:

$$\sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii} = \text{tr}(\mathbf{A}) \quad (1.1.4.1)$$

A matrix is symmetric if $\mathbf{A}^\top = \mathbf{A}$ and skew-symmetric if $\mathbf{A}^\top = -\mathbf{A}$ [cite: 1].

1.1.5 Cholesky Decomposition

- Factorization: A symmetric positive definite matrix $\mathbf{A}$ can be factorized as $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$ where $\mathbf{L}$ is lower triangular[cite: 1].
- 2×2 Decomposition Form:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix} = \begin{pmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{pmatrix} \begin{pmatrix} l_{11} & l_{21} \\ 0 & l_{22} \end{pmatrix} = \begin{pmatrix} l_{11}^2 & l_{11}l_{21} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 \end{pmatrix} \quad (1.1.5.1)$$

1.1.6 Bayes' Theorem in Vector Form

- Vector Formulation: Given prior vector $\pi = (\text{Pr}(B_1) \dots \text{Pr}(B_n))^\top$ and likelihood vector $\mathbf{l} = (\text{Pr}(E | B_1) \dots \text{Pr}(E | B_n))^\top$ [cite: 1]:

$$\text{Pr}(E) = \pi^\top \mathbf{l} \quad (1.1.6.1)$$

$$\text{Pr}(B_i | E) = \frac{\pi_i l_i}{\pi^\top \mathbf{l}} \quad (1.1.6.2)$$

1.1.7 Minimization via Completing the Square

- Inequality and Equality Condition: For real a and b [cite: 1]:

$$ab - b^2 = \frac{a^2}{4} - \left(b - \frac{a}{2}\right)^2 \leq \frac{a^2}{4} \quad (1.1.7.1)$$

with equality holding if and only if $b = \frac{a}{2}$ [cite: 1].

- Zero Integral Property: If $g(x) \geq 0$ is continuous on $[a, b]$ and $\int_a^b g(x) dx = 0$, then $g(x) = 0$ on $[a, b]$ [cite: 1].

1.1.8 Laplace Transform and State-Space Form

- Transform Operational Properties:

$$y'(t) \xleftrightarrow{\mathcal{L}} sY(s) - y(0) \quad (1.1.8.1)$$

$$y''(t) \xleftrightarrow{\mathcal{L}} s^2Y(s) - sy(0) - y'(0) \quad (1.1.8.2)$$

$$e^{at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s-a} \quad (1.1.8.3)$$

- State-Space Representation: Writing $\mathbf{z} = \begin{pmatrix} y \\ y' \end{pmatrix}^\top$, the ODE $y'' + py' + qy = 0$ becomes $\mathbf{z}' = \mathbf{A}\mathbf{z}$ where[cite: 1]:

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix} \quad (1.1.8.4)$$

The exponents of the solution are the eigenvalues of $\mathbf{A}$ [cite: 1].

1.1.9 Newton-Raphson Method

- First-Order Recurrence: For a differentiable function $f(x)$, the root is iteratively approximated by[cite: 1]:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (1.1.9.1)$$

1.1.10 Polynomial Interpolation (Vandermonde Matrix)

- Matrix Form: The polynomial $P_2(x) = c_0 + c_1x + c_2x^2$ through points (x_i, y_i) for $i = 1, 2, 3$ satisfies[cite: 1]:

$$\begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \quad (1.1.10.1)$$

1.1.11 Equilibrium of Rigid Bodies in Vector Form

- Force and Moment Equilibrium: For a body in equilibrium under forces $\mathbf{F}_i$ acting at points $\mathbf{r}_i$ [cite: 1]:

$$\sum_i \mathbf{F}_i = \mathbf{0}, \quad \sum_i (\mathbf{r}_i - \mathbf{O}) \times \mathbf{F}_i = \mathbf{0} \quad (1.1.11.1)$$

where in the 2D plane, $\mathbf{r} \times \mathbf{F} = r_x F_y - r_y F_x$ [cite: 1].

- Bending Moment and Friction: The bending moment at section P is $M(P) = \sum_{\text{one side}} (\mathbf{r}_i - \mathbf{P}) \times \mathbf{F}_i$ [cite: 1]. Limiting friction force is $F = \mu N$ [cite: 1].

1.1.12 Direct Stiffness Method

- Member and Global Stiffness Formulation: For an axial member of length L , area A , modulus E , and direction vector $\mathbf{m}$, the nodal stiffness is $\mathbf{k} = \frac{AE}{L} \mathbf{m} \mathbf{m}^\top$ and member stiffness matrix is $\begin{pmatrix} \mathbf{k} & -\mathbf{k} \\ -\mathbf{k} & \mathbf{k} \end{pmatrix}$ [cite: 1]. The global system is $\mathbf{K} \mathbf{u} = \mathbf{F}$ [cite: 1].
- Thermal Load Equivalence: A temperature rise ΔT with expansion coefficient α produces equivalent nodal loads of $AE\alpha\Delta T$ at member ends directed outward[cite: 1].

1.1.13 Cellular Update Rule in Matrix Form

- Matrix Formulation: For N bulbs in a line with state vector $\mathbf{x}_k \in \{0, 1\}^N$, the number of ON neighbours is given by $s_k = \mathbf{A} \mathbf{x}_k$ with[cite: 1]:

$$a_{ij} = \begin{cases} 1, & |i - j| = 1 \\ 0, & \text{otherwise} \end{cases} \quad (1.1.13.1)$$

1.1.14 Surveying and Traffic Relations

- Reduced Level: With height of instrument HI and staff reading h , $RL = HI - h$ [cite: 1].
- Space Mean Speed: For n vehicles covering distance d in times $\mathbf{t} = (t_1 \ \dots \ t_n)^\top$ [cite: 1]:

$$v_s = \frac{nd}{\mathbf{1}^\top \mathbf{t}} = \frac{n}{\sum_i \frac{1}{v_i}} \quad (1.1.14.1)$$

which is the harmonic mean of the spot speeds $v_i = d/t_i$ [cite: 1].

B. Problems

1.2.1 Intersection of Conics and Common Chord

- Conic in Vector Form: Every second-degree curve $ax^2 + 2bxy + cy^2 + 2dx + 2ey + f = 0$ can be written in matrix-vector form as:

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0, \quad \mathbf{V} = \begin{pmatrix} a & b \\ b & c \end{pmatrix} = \mathbf{V}^\top, \quad \mathbf{u} = \begin{pmatrix} d \\ e \end{pmatrix} \quad (1.2.1.1)$$

The conic is a parabola if $|\mathbf{V}| = 0$, an ellipse if $|\mathbf{V}| > 0$, and a hyperbola if $|\mathbf{V}| < 0$. For a circle with centre $\mathbf{c}$ and radius r , $\|\mathbf{x} - \mathbf{c}\|^2 = r^2$ gives $\mathbf{V} = \mathbf{I}$, $\mathbf{u} = -\mathbf{c}$, and $f = \|\mathbf{c}\|^2 - r^2$.

- Line-Conic Intersection: The points of intersection of a line $L : \mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ ($\kappa \in \mathbb{R}$) with a conic section $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$ are given by:

$$\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m} \quad (1.2.1.2)$$

where the parameter values κ_i are:

$$\kappa_i = \frac{-\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h}) (\mathbf{m}^\top \mathbf{V} \mathbf{m})}}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \quad (1.2.1.3)$$

with $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V} \mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f$. The line meets the conic at two, one, or no real points according as the discriminant

$$\Delta = [\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h}) (\mathbf{m}^\top \mathbf{V} \mathbf{m}) \quad (1.2.1.4)$$

is positive, zero, or negative.

- Derivation of the Intersection Parameter: Substituting $\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ into the conic equation and using symmetry $\mathbf{V} = \mathbf{V}^\top$:

$$(\mathbf{h} + \kappa \mathbf{m})^\top \mathbf{V} (\mathbf{h} + \kappa \mathbf{m}) + 2\mathbf{u}^\top (\mathbf{h} + \kappa \mathbf{m}) + f = 0 \quad (1.2.1.5)$$

$$\kappa^2 (\mathbf{m}^\top \mathbf{V} \mathbf{m}) + 2\kappa \mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) + g(\mathbf{h}) = 0 \quad (1.2.1.6)$$

When $\Delta = 0$, the two roots coincide and the line is tangent to the conic.

- Pencil of Conics: The intersection of two conics with parameters $(\mathbf{V}_1, \mathbf{u}_1, f_1)$ and $(\mathbf{V}_2, \mathbf{u}_2, f_2)$ lies on every member of the pencil:

$$\mathbf{x}^\top (\mathbf{V}_1 + \mu \mathbf{V}_2) \mathbf{x} + 2(\mathbf{u}_1 + \mu \mathbf{u}_2)^\top \mathbf{x} + (f_1 + \mu f_2) = 0 \quad (1.2.1.7)$$

- Degenerate Conic and Common Chord: The pencil represents a pair of straight lines (a degenerate conic) if and only if:

$$\begin{vmatrix} \mathbf{V}_1 + \mu \mathbf{V}_2 & \mathbf{u}_1 + \mu \mathbf{u}_2 \\ (\mathbf{u}_1 + \mu \mathbf{u}_2)^\top & f_1 + \mu f_2 \end{vmatrix} = 0 \quad (1.2.1.8)$$

If $\mathbf{V}_1 = \mathbf{V}_2$ (e.g., two circles), choosing $\mu = -1$ removes the second-degree terms, leaving the common chord (radical axis):

$$2(\mathbf{u}_1 - \mathbf{u}_2)^\top \mathbf{x} + (f_1 - f_2) = 0 \quad (1.2.1.9)$$

1.2.2 Line Intersection and Angle Between Vectors

- Forms of a Line: A line through point $\mathbf{A}$ with direction vector $\mathbf{m}$ can be written in parametric form or normal form using normal vector $\mathbf{n}$ ($\mathbf{n}^\top \mathbf{m} = 0$):

$$\mathbf{x} = \mathbf{A} + \kappa \mathbf{m}, \quad \mathbf{n}^\top (\mathbf{x} - \mathbf{A}) = 0 \quad (1.2.2.1)$$

In 2D, if $\mathbf{m} = \begin{pmatrix} m_1 \\ m_2 \end{pmatrix}$, then $\mathbf{n} = \begin{pmatrix} -m_2 \\ m_1 \end{pmatrix}$.

- Line Intersection and Solvability: The lines $\mathbf{x} = \mathbf{A} + \kappa_1 \mathbf{m}_1$ and $\mathbf{x} = \mathbf{B} + \kappa_2 \mathbf{m}_2$ meet where:

$$\begin{pmatrix} \mathbf{m}_1 & -\mathbf{m}_2 \end{pmatrix} \begin{pmatrix} \kappa_1 \\ \kappa_2 \end{pmatrix} = \mathbf{B} - \mathbf{A} \quad (1.2.2.2)$$

This system has a unique solution if and only if $|\mathbf{m}_1 \quad \mathbf{m}_2| \neq 0$. If the determinant is zero, the lines are parallel or coincident.

- Angle Between Vectors: The angle θ between two vectors $\mathbf{a}$ and $\mathbf{b}$ is given by:

$$\cos \theta = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \quad (1.2.2.3)$$

Two vectors are orthogonal if and only if $\mathbf{a}^\top \mathbf{b} = 0$.

- Circle Representation and Chords: A point on a circle of radius r centered at the origin is $\mathbf{x} = r \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix}$ with $\mathbf{e}_\alpha \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix}$. The chord length between $\mathbf{x}_\alpha = r \mathbf{e}_\alpha$ and $\mathbf{x}_\beta = r \mathbf{e}_\beta$ satisfies:

$$\mathbf{e}_\alpha^\top \mathbf{e}_\beta = \cos(\alpha - \beta) \quad (1.2.2.4)$$

$$\|\mathbf{x}_\alpha - \mathbf{x}_\beta\|^2 = 2r^2 (1 - \mathbf{e}_\alpha^\top \mathbf{e}_\beta) = 4r^2 \sin^2 \frac{\alpha - \beta}{2} \quad (1.2.2.5)$$

1.2.3 System of Linear Equations and Rouché–Capelli Theorem

- Consistency and Solvability: A system

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{A} \in \mathbb{R}^{m \times n} \quad (1.2.3.1)$$

is consistent if and only if

$$\text{rank}(\mathbf{A}) = \text{rank}([\mathbf{A} \mid \mathbf{b}]) = r \quad (1.2.3.2)$$

If $r = n$, the solution is unique; if $r < n$, there exist infinitely many solutions with $n - r$ free variables. If $\mathbf{A} \in \mathbb{R}^{n \times (n-1)}$, the system is consistent only if

$$|\mathbf{A} \quad \mathbf{b}| = 0 \quad (1.2.3.3)$$

- Homogeneous System and Zero Norm Property: For $\mathbf{A} \in \mathbb{R}^{2 \times 3}$ with rank 2, the solution of $\mathbf{A}\mathbf{x} = \mathbf{0}$ is given by the cross product of its rows:

$$\mathbf{x} = \kappa (\mathbf{a}_1 \times \mathbf{a}_2) \quad (1.2.3.4)$$

For any vector $\mathbf{v} \in \mathbb{R}^n$,

$$\|\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{v} = \sum_i v_i^2 = 0 \iff \mathbf{v} = \mathbf{0} \quad (1.2.3.5)$$

1.2.4 Relation Between Eigenvalues and Trace of a Matrix

- **Characteristic Polynomial and Trace Relation:** The characteristic polynomial of $\mathbf{A} \in \mathbb{R}^{n \times n}$ is

$$P(\lambda) = |\lambda \mathbf{I} - \mathbf{A}| \quad (1.2.4.1)$$

The sum of all eigenvalues equals the trace of the matrix:

$$\sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii} = \text{tr}(\mathbf{A}) \quad (1.2.4.2)$$

A matrix is symmetric if $\mathbf{A}^\top = \mathbf{A}$ and skew-symmetric if $\mathbf{A}^\top = -\mathbf{A}$.

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- **Factorization:** A symmetric positive definite matrix $\mathbf{A}$ can be factorized as $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$ where $\mathbf{L}$ is lower triangular.
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$$\begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix} = \begin{pmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{pmatrix} \begin{pmatrix} l_{11} & l_{21} \\ 0 & l_{22} \end{pmatrix} = \begin{pmatrix} l_{11}^2 & l_{11}l_{21} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 \end{pmatrix} \quad (1.2.5.1)$$

1.2.6 Bayes' Theorem in Vector Form

- **Vector Formulation:** Given prior vector $\pi = (\Pr(B_1) \ \dots \ \Pr(B_n))^\top$ and likelihood vector $\mathbf{l} = (\Pr(E | B_1) \ \dots \ \Pr(E | B_n))^\top$:

$$\Pr(E) = \pi^\top \mathbf{l} \quad (1.2.6.1)$$

$$\Pr(B_i | E) = \frac{\pi_i l_i}{\pi^\top \mathbf{l}} \quad (1.2.6.2)$$

1.2.7 Minimization via Completing the Square

- **Inequality and Equality Condition:** For real a and b :

$$ab - b^2 = \frac{a^2}{4} - \left(b - \frac{a}{2}\right)^2 \leq \frac{a^2}{4} \quad (1.2.7.1)$$

with equality holding if and only if $b = \frac{a}{2}$.

- **Zero Integral Property:** If $g(x) \geq 0$ is continuous on $[a, b]$ and $\int_a^b g(x) dx = 0$, then $g(x) = 0$ on $[a, b]$.

1.2.8 Laplace Transform and State-Space Form

- **Transform Operational Properties:**

$$y'(t) \xrightarrow{\mathcal{L}} sY(s) - y(0) \quad (1.2.8.1)$$

$$y''(t) \xrightarrow{\mathcal{L}} s^2Y(s) - sy(0) - y'(0) \quad (1.2.8.2)$$

$$e^{at}u(t) \xrightarrow{\mathcal{L}} \frac{1}{s-a} \quad (1.2.8.3)$$

- **State-Space Representation:** Writing $\mathbf{z} = \begin{pmatrix} y \\ y' \end{pmatrix}^\top$, the ODE $y'' + py' + qy = 0$

becomes $\mathbf{z}' = \mathbf{A}\mathbf{z}$ where:

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix} \quad (1.2.8.4)$$

The exponents of the solution are the eigenvalues of $\mathbf{A}$.

1.2.9 Newton-Raphson Method

- First-Order Recurrence: For a differentiable function $f(x)$, the root is iteratively approximated by:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (1.2.9.1)$$

1.2.10 Polynomial Interpolation (Vandermonde Matrix)

- Matrix Form: The polynomial $P_2(x) = c_0 + c_1x + c_2x^2$ through points (x_i, y_i) for $i = 1, 2, 3$ satisfies:

$$\begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \quad (1.2.10.1)$$

1.2.11 Equilibrium of Rigid Bodies in Vector Form

- Force and Moment Equilibrium: For a body in equilibrium under forces $\mathbf{F}_i$ acting at points $\mathbf{r}_i$:

$$\sum_i \mathbf{F}_i = \mathbf{0}, \quad \sum_i (\mathbf{r}_i - \mathbf{O}) \times \mathbf{F}_i = \mathbf{0} \quad (1.2.11.1)$$

where in the 2D plane, $\mathbf{r} \times \mathbf{F} = r_x F_y - r_y F_x$.

- Bending Moment and Friction: The bending moment at section P is

$$M(P) = \sum_{\text{one side}} (\mathbf{r}_i - \mathbf{P}) \times \mathbf{F}_i \quad (1.2.11.2)$$

Limiting friction force is $F = \mu N$.

1.2.12 Direct Stiffness Method

- Member and Global Stiffness Formulation: For an axial member of length L , area A , modulus E , and direction vector $\mathbf{m}$, the nodal stiffness is

$$\mathbf{k} = \frac{AE}{L} \mathbf{m} \mathbf{m}^\top \quad (1.2.12.1)$$

and the member stiffness matrix is $\begin{pmatrix} \mathbf{k} & -\mathbf{k} \\ -\mathbf{k} & \mathbf{k} \end{pmatrix}$. The global system is

$$\mathbf{K} \mathbf{u} = \mathbf{F} \quad (1.2.12.2)$$

- Thermal Load Equivalence: A temperature rise ΔT with expansion coefficient α produces equivalent nodal loads of $AE\alpha\Delta T$ at member ends directed outward.

1.2.13 Cellular Update Rule in Matrix Form

- Matrix Formulation: For N bulbs in a line with state vector $\mathbf{x}_k \in \{0, 1\}^N$, the

number of ON neighbours is given by $\mathbf{s}_k = \mathbf{A}\mathbf{x}_k$ with:

$$a_{ij} = \begin{cases} 1, & |i - j| = 1 \\ 0, & \text{otherwise} \end{cases} \quad (1.2.13.1)$$

1.2.14 Surveying and Traffic Relations

- Reduced Level: With height of instrument HI and staff reading h ,

$$RL = HI - h \quad (1.2.14.1)$$

- Space Mean Speed: For n vehicles covering distance d in times $\mathbf{t} = (t_1 \ \dots \ t_n)^\top$:

$$v_s = \frac{nd}{\mathbf{1}^\top \mathbf{t}} = \frac{n}{\sum_i \frac{1}{v_i}} \quad (1.2.14.2)$$

which is the harmonic mean of the spot speeds $v_i = d/t_i$.